1) High-level description of gradient descent idea
2) Bisection and golden section search for single-dim optimization
3) Penalty function method, barrier function method
4) Taylor's theorem and its implication
5) Definition of a convex function
6) First-order characterization of convexity
7) Convergence of gradient descent with fixed step size for functions with bounded gradients (Preliminary analysis)
8) Convergence of gradient descent with fixed step size for smooth convex functions
9) Convex sets

Gradient Descent

Start with some initial point x^0, and generate a sequence of points $x^1, x^2, x^3, \ldots$

$$x^{k+1} = x^k - \gamma \cdot \nabla f(x^k)$$

Homework 3, Question 2

a) Consider the single-dimensional function $f(x) = x^2$. Show that this function satisfies the inequality $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$. for all $x, y \in \mathbb{R}$ and $\lambda \in [0,1]$. Thus, $f : \mathbb{R} \to \mathbb{R}$ is convex.

We want to show $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$

We want to show $(\lambda x + (1-\lambda) y)^2 \leq \lambda x^2 + (1-\lambda) y^2$

Is it true that

$$(\lambda x + (1-\lambda) y)^2 \leq \lambda x^2 + (1-\lambda) y^2 \qquad (\ast)$$

Is it true that

$$\lambda^2 x^2 + (1-\lambda)^2 y^2 + 2\lambda (1-\lambda) xy \leq \lambda x^2 + (1-\lambda) y^2$$

Is it true that

$$2\lambda (1-\lambda) xy \leq x^2 (\lambda - \lambda^2) + y^2 ((1-\lambda) - (1-\lambda)^2)$$

Is it true that

$$2 \cancel{\lambda} \cancel{(1-\lambda)} xy \leq x^2 \cancel{\lambda} \cancel{(1-\lambda)} + y^2 \cancel{(1-\lambda)} \cancel{\lambda}$$

Is it true that

$$0 \leq x^2 + y^2 - 2xy$$

Is it true that $0 \leq (x-y)^2$ Yes! Done!

Homework 3 . Question 2

b) Repeat Part (a) for the single-dimensional function $f(x) = (x-a)^2$, where a is a fixed number.

we want to check $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$

Is it true that

$$(\lambda x + (1-\lambda) y - a)^2 \leq \lambda (x-a)^2 + (1-\lambda) (y-a)^2$$

Is it true that

$$(\lambda x + (1-\lambda) y - \lambda a - (1-\lambda) a)^2 \leq \lambda (x-a)^2 + (1-\lambda)(y-a)^2$$

Is it true that

$$(\lambda (x-a) + (1-\lambda)(y-a))^2 \leq \lambda (x-a)^2 + (1-\lambda)(y-a)^2 \quad (*)$$

$(*)$ and $(\dagger)$ are the same after replacing x and y with $(x-a)$ and $(y-a)$,

so do the same steps in Part (a)

Homework 3, Question 2

c) For an n-dimensional vector $\boldsymbol{x} = (x_1, \ldots, x_n)$, consider its squared distance from the fixed point $\boldsymbol{a}$ given by $\|\boldsymbol{x} - \boldsymbol{a}\|^2 = \sum_{i=1}^{n} (x_i - a_i)^2$. Show that the n-dimensional function $f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{a}\|^2$ is convex.

$$f(x) = \sum_{i=1}^{n} (x_i - a_i)^2$$

n-dim vector

We want to show

$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$$

$$\lambda x + (1-\lambda) y = \lambda \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + (1-\lambda) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 + (1-\lambda) y_1 \\ \lambda x_2 + (1-\lambda) y_2 \\ \vdots \\ \lambda x_n + (1-\lambda) y_n \end{pmatrix}$$

$$f(\lambda x + (1-\lambda)y) = \sum_{i=1}^{n} (\lambda x_i + (1-\lambda)y_i - a_i)^2$$

$$f(x) = \sum_{i=1}^{n} (x_i - a_i)^2 \qquad f(y) = \sum_{i=1}^{n} (y_i - a_i)^2$$

We want to show

$$\sum_{i=1}^{n} (\lambda x_i + (1-\lambda)y_i - a_i)^2 \leq \lambda \sum_{i=1}^{n} (x_i - a_i)^2 + (1-\lambda) \sum_{i=1}^{n} (y_i - a_i)^2$$

We want to show

$$\sum_{i=1}^{n} (\lambda x_i + (1-\lambda)y_i - a_i)^2 \leq \sum_{i=1}^{n} \left(\lambda (x_i - a_i)^2 + (1-\lambda)(y_i - a_i)^2\right)$$

By Part (b), we know

$$(\lambda x_i + (1-\lambda)y_i - a_i)^2 \leq \lambda (x_i - a_i)^2 + (1-\lambda)(y_i - a_i)^2$$

for all $i = 1, \ldots, n$

Thus, adding these inequalities for $i = 1, \ldots, n$, we get

$$\sum_{i=1}^{n} (\lambda x_i + (1-\lambda)y_i - a_i)^2 \leq \sum_{i=1}^{n} \left(\lambda (x_i - a_i)^2 + (1-\lambda)(y_i - a_i)^2\right)$$

done!

Question 3

We say that the function $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex with parameter μ if we have

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla^t f(\boldsymbol{x})\,(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mu}{2}\,\|\boldsymbol{x} - \boldsymbol{y}\|^2$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Thus, a strongly convex function is lower bounded by a local quadratic approximation.

Let the sequence of points $\boldsymbol{x}^0, \boldsymbol{x}^1, \boldsymbol{x}^2, \ldots$ be generated by gradient descent with constant step size, so $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \gamma\,\nabla f(\boldsymbol{x}^k)$. Using $\boldsymbol{x}^*$ to denote a minimum of f, show that if f is strongly convex with parameter μ, then we have the bound

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\|^2 \leq (1 - \mu\,\gamma)\,\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2 + 2\,\gamma\,(f(\boldsymbol{x}^*) - f(\boldsymbol{x}^k)) + \gamma^2\,\|\nabla f(\boldsymbol{x}^k)\|^2.$$

To interpret the inequality above, because $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}^k)$, the inequality implies that we have $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\|^2 \leq (1 - \mu\,\gamma)\,\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2 + \gamma^2\,\|\nabla f(\boldsymbol{x}^k)\|^2$. Thus, as the gradient of the f at $\boldsymbol{x}^k$ becomes small, we get $\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\|^2 \leq (1 - \mu\,\gamma)\,\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$, which means that gradient descent gets us closer to the minimum at every iteration, as long as we choose the step size small enough so that $1 - \mu\,\gamma \geq 0$.

(Hint: To show the desired inequality, write the definition of a strongly convex function with $\boldsymbol{x} = \boldsymbol{x}^k$ and $\boldsymbol{y} = \boldsymbol{x}^*$. Collect $f(\boldsymbol{x}^*) - f(\boldsymbol{x}^k)$ to one side of the inequality and collect all other terms to the other side. Use the fact that $\nabla f(\boldsymbol{x}^k) = \frac{1}{\gamma}\,(\boldsymbol{x}^k - \boldsymbol{x}^{k+1})$, as well as the equality $\boldsymbol{x}^t\,\boldsymbol{y} = \frac{1}{2}(\|\boldsymbol{x} + \boldsymbol{y}\|^2 - \|\boldsymbol{x}\|^2 - \|\boldsymbol{y}\|^2)$, which is very similar to the equality we showed in class.)

$$x^{k+1} = x^k - \gamma \nabla f(x^k)$$

$$f(x^*) \geq f(x^k) + \underbrace{\nabla^t f(x^k)}_{\frac{1}{\gamma}\cdot(x^k - x^{k+1})}(x^* - x^k) + \frac{\mu}{2}\|x^k - x^*\|^2 \Rightarrow$$

$$f(x^*) \geq f(x^k) + \frac{1}{\gamma}(x^k - x^{k+1})^t (x^* - x^k) + \frac{\mu}{2}\|x^k - x^*\|^2 \quad (*)$$

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2x^t y \Rightarrow x^t y = \frac{1}{2}\left(\|x+y\|^2 - \|x\|^2 - \|y\|^2\right)$$

Thus, from $(*)$ we get

$$f(x^*) \geq f(x^k) + \frac{1}{2\gamma} \cdot \left(\cancel{\|x^k - x^{k+1} + x^* - x^k\|^2} - \|x^k - x^{k+1}\|^2 - \|x^* - x^k\|^2 \right)$$

$$+ \frac{\mu}{2} \|x^k - x^*\|^2 \quad \Longrightarrow$$

$$f(x^*) \geq f(x^k) + \frac{1}{2\gamma} \left(\|x^* - x^{k+1}\|^2 - \boxed{\|x^k - x^{k+1}\|^2} - \|x^* - x^k\|^2 \right)$$

$$+ \frac{\mu}{2} \|x^k - x^*\|^2 \quad \Longrightarrow$$

$$2\gamma \left(f(x^*) - f(x^k) \right) \geq \|x^* - x^{k+1}\|^2 - \boxed{\gamma^2 \|\nabla f(x^k)\|^2}$$

$$- \|x^* - x^k\|^2 + \mu\gamma \|x^k - x^*\|^2$$

$\Longrightarrow$

$$\|x^{k+1} - x^*\|^2 \leq 2\gamma \left(f(x^*) - f(x^k) \right) + \gamma^2 \|\nabla f(x^k)\|^2$$

$$+ (1 - \mu\gamma) \|x^k - x^*\|^2$$

Practice Problem

$$\left(\max\{g(x), 0\}\right)^2$$
$$g(x) \le 0$$

$$\min_{x} \quad e^{x_1} + 100\,(x_2+2)^2$$

$$x_1 + x_2 \geq 10 \qquad (10 - x_1 - x_2 \le 0)$$
$$x_1 \geq 7 \qquad (7 - x_1 \le 0)$$

We want to solve this problem using penalty function method.

Use penalty parameters $\theta_1 = \theta_2 = 100$ for the constraints.

Start from initial point $x^0 = (x_1^0, x_2^0) = (5, 6)$

Do one iteration of gradient descent with step size 0.01 to minimize the penalized objective function.

Penalized objective function

$$e^{x_1} + 100 \cdot (x_2+2)^2 + \theta_1 \left(\max\{10 - x_1 - x_2, 0\}\right)^2$$
$$+ \theta_2 \left(\max\{7 - x_1, 0\}\right)^2 = h(x)$$

$$\downarrow$$

$$(x_1, x_2)$$

We want to compute $\nabla h(5,6)$.

Around a small neighborhood of $(5,6)$, $h(x)$ behaves like

$$e^{x_1} + 100 \cdot (x_2+2)^2 + 100\,(7 - x_1)^2$$

Thus, around a small neighborhood of $(5,6)$, $\nabla h(x)$ is

$$\nabla h(x) = \begin{pmatrix} e^{x_1} - 200 \cdot (7 - x_1) \\ 200 \cdot (x_2 + 2) \end{pmatrix}$$

$$\nabla h(x) = \begin{pmatrix} e^{5} - 200\,(7-5) \\ 200\,(6+2) \end{pmatrix} = \begin{pmatrix} -251{,}57 \\ 1600 \end{pmatrix}$$

Thus:

$$\begin{pmatrix} x_1^{k+1} \\ x_2^{k+1} \end{pmatrix} = \overbrace{\begin{pmatrix} 5 \\ 6 \end{pmatrix}}^{x^k} - 0{,}01 \overbrace{\begin{pmatrix} -251{,}57 \\ 1600 \end{pmatrix}}^{\nabla h(x^k)}$$

$$= \begin{pmatrix} 5 + 2{,}5157 \\ 6 - 16 \end{pmatrix} = \begin{pmatrix} 7{,}5157 \\ -10 \end{pmatrix}$$

If we were to do one more iteration of gradient descent starting at $(7{,}5157, -10)$, then around a small neighborhood $(7{,}5157, -10)$, the penalized objective function behaves like

$$e^{x_1} + 100 \cdot (x_2 + 2)^2 + 100 \cdot (10 - x_1 - x_2)^2$$